

Structure Overview

Data Base :

school

Main Tables:

- Students
- Teachers
- Classes
- Subjects
- Fees
- Contact_Info

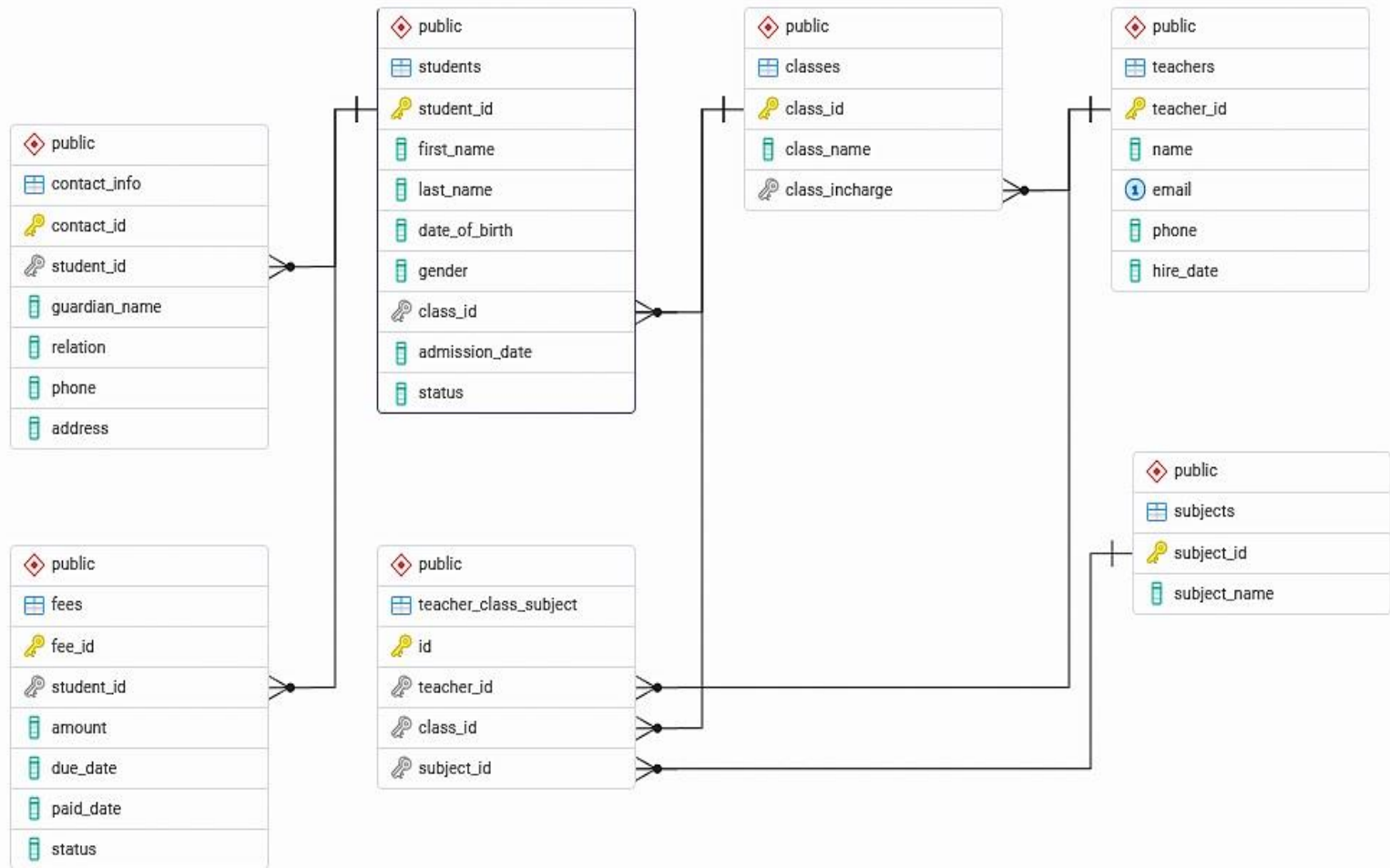
Supporting Table:

- Teacher_Class_Subject (mapping)
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Relationships Summary

- **Students → Classes**
 - Many students belong to one class
- **Teachers → Classes**
 - One teacher can be class teacher (1–1 optional)
- **Teachers ↔ Classes (via Subjects)**
 - Many-to-many via mapping table
- **Students → Fees**
 - One-to-many
- **Students → Contact Info**
 - One-to-many

ER Diagram



Users and Permissions

Users	Passwords	Scope	Permissions
yousuf	yousuf	school DB	Full control on 7 tables
talha	talha	school DB	SELECT, INSERT, UPDATE
umar	umar	school DB	SELECT only

Views

View Name	Purpose
student_complete_details	Full student profile
class_subject_teacher	Teaching + subjects +
student_fee_details	Fee tracking with student info
student_contact_details	Student + Guardian info
class_overview	Student count per class

Triggers

First we created 2 more tables:

deleted_student_log table

teacher_log table

then we create 2 functions and attach them to 2 triggers.

Functioning of triggers:

As the name says, record of deleted students will be automatically kept in deleted_student_log table.

And any change in teachers table will be recorded in the teacher_log table by the triggers.

<https://github.com/Muhammad-Yousuf-Khan/DBA-Project.git>
